Native Profile Budget Frontiers for Finite Twin-Prime Shells

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29 August 2026

Abstract

The profile-angle ladder of TPC-298 measures when a literal source family becomes geometrically expressive, but it does not measure the source norm spent by that expression. We introduce the native profile budget frontier

$$\mathcal{B}_{k,\tau}(b) = \min\{c^{\mathsf{T}} M_k c : \|V_k c - b\|_2 \le \tau \|b\|_2\}, \qquad M_k = U_k^{\mathsf{T}} U_k, \quad V_k = A^{\mathsf{T}} U_k,$$

and prove its exact ridge/KKT representation, feasibility criterion, and monotonicity under nested prefixes. On the inherited 18-row, 1,380-edge literal grid, the weighted sign target at normalized RMS 1/2 requires more than $9 \cdot 10^{-5} \|\beta\|_2^2$ on every row, more than $10^{-3} \|\beta\|_2^2$ on 14 rows, and more than the latter threshold on 11 rows even after the full available 17-profile prefix is used. The all-positive control stays below $10^{-4} \|\beta\|_2^2$ on all 18 rows, and the weighted/positive budget ratio exceeds 20 on every row. These are finite restricted-profile diagnostics: no growing budget theorem, arithmetic L^2 estimate, Gate-B endpoint, or twin-prime conclusion is claimed.

1 Position on the route

The current finite prime-shell line separates three questions that are easy to conflate. TPC-295 showed that an unrestricted rational source correlation map can be onto on the finite grid [4]. TPC-296 computed the least-norm cost in that unrestricted ambient source space and found cheap witnesses, while a frozen native ray remained poorly aligned [1]. TPC-297 and TPC-298 replaced the ray by literal cutoff profiles and measured the resulting image angle and dimension ladder [3, 2]. The present paper asks the next minimal question: when a profile prefix reaches a target tolerance, what is the norm of the actual source vector $U_k c$?

The distinction matters. A small Euclidean norm of the coefficient vector does not imply a small source norm when the profile columns are correlated. Conversely, a finite full-rank image can hide a large source budget behind the final surjectivity point. We therefore keep the source Gram matrix in the optimization throughout.

2 Finite physical map and literal source family

Let I be the frozen integer interval and let $S = \{q : Q < q \le 2Q,\ q \text{ prime}\}$. For a prime q, height H, and kernel exponent s, the registered physical column is

$$(g_q)_u = \sum_{\substack{t\in I\\ t\neq u}} q K_{H,s}(u-t) B_q(u,t)\beta_t, \qquad K_{H,s}(v) = \frac{H^{2s}}{(H^2+v^2)^s},$$

$$B_q(u,t) = \mathbf{1}_{q\nmid u}\mathbf{1}_{q\nmid t}\left(\mathbf{1}_{u\equiv t \pmod q} - \frac{1}{q-1}\right).$$

Here β is the frozen source profile used to register the physical operator. Put $A = [g_q]_{q\in S}$.

For the ordered literal cutoff list

$$Z = (3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61),$$

define

$$\lambda(t) = \begin{cases} 1/j, & t = p^j \text{ for a prime } p, \\ 0, & \text{otherwise,} \end{cases} \qquad \beta_z(t) = \lambda(t) - \sum_{\substack{d \le z \\ d \mid t}} \mu(d).$$

Let $U_k = [u_{z_1}, \ldots, u_{z_k}]$, where $(u_z)_t = \beta_z(t)$, and define

$$V_k = A^{\mathsf{T}} U_k, \qquad M_k = U_k^{\mathsf{T}} U_k.$$

Thus a coefficient vector c produces target $V_k c$ and has actual native source budget

$$\|U_k c\|_2^2 = c^{\mathsf{T}} M_k c.$$

The source directions are fixed before inspecting the target. The weighted minimum, unit-edge max-cut, and all-positive vectors below are finite target controls inherited from the earlier route stages.

Remark 1. *The word literal refers to the displayed finite cutoff formula. It does not assert that this 17-element family is complete, uniform in the shell scale, or equivalent to the unrestricted arithmetic source class.*

3 Exact budget frontier

For a nonzero target $b \in \mathbb{R}^{|S|}$ and $0 \le \tau < 1$, define

$$\mathcal{B}_{k,\tau}(b) = \min \left\{ c^{\mathsf{T}} M_k c : \|V_k c - b\|_2 \le \tau \|b\|_2 \right\}.$$

If the constraint is infeasible, the value is $+\infty$. The next theorem is the source-budget analogue of the projection identity in TPC-298.

Theorem 1 (Finite native budget frontier)**.** *Suppose $M = U^{\mathsf{T}} U$ is positive definite and $V = A^{\mathsf{T}} U$. Set $R = \tau \|b\|_2$ with $0 \le R < \|b\|_2$. If*

$$\operatorname{dist}(b, \operatorname{col} V) \le R,$$

then (3) has a unique minimizer. If the least-squares distance is strictly smaller than R, there is a multiplier $\lambda > 0$ such that

$$c_\lambda = (V^{\mathsf{T}} V + \lambda M)^{-1} V^{\mathsf{T}} b, \qquad \|V c_\lambda - b\|_2 = R,$$

and $\mathcal{B}_{k,\tau}(b) = c_\lambda^{\mathsf{T}} M c_\lambda$. At the least-squares boundary one takes $\lambda = 0$. If the displayed distance condition fails, the frontier is infeasible.

Proof. Let $y = M^{1/2} c$ and $W = V M^{-1/2}$. The problem becomes

$$\min_y \|y\|_2^2 \quad \text{subject to} \quad \|Wy - b\|_2^2 \le R^2.$$

The feasible set is closed and convex, and the objective is strictly convex, so a feasible problem has a unique minimizer. When the least-squares distance is strictly below R, the least-squares point is a strict feasible point. The convex KKT conditions are therefore necessary and sufficient. Stationarity for the Lagrangian with multiplier λ is

$$y + \lambda W^{\mathsf{T}} (Wy - b) = 0.$$

Undoing the change of variables gives $(V^{\mathsf{T}} V + \lambda M)c = V^{\mathsf{T}} b$. Complementary slackness makes the target constraint active, and the objective becomes $c^{\mathsf{T}} M c$. If R is below the projection distance there is no feasible point; if equality holds, the least-squares minimizer is already optimal and $\lambda = 0$. □

Proposition 1 (Ridge path and budget feasibility)**.** *For $\lambda > 0$, $V^{\mathsf{T}} V + \lambda M$ is positive definite. Along the ridge path c_λ, the residual increases continuously toward $\|b\|_2$ while the source norm decreases toward zero, subject to the usual omission of components invisible to V. Consequently a source budget B reaches tolerance τ if and only if $B \ge \mathcal{B}_{k,\tau}(b)$.*

Proof. After whitening, take an SVD of W. If σ_j are its nonzero singular values and a_j are the target coordinates in the corresponding left singular vectors, then

$$\|W y_\lambda - b\|_2^2 = \|b_\perp\|_2^2 + \sum_j \left(\frac{\lambda}{\sigma_j^2 + \lambda} \right)^2 a_j^2,$$

$$\|y_\lambda\|_2^2 = \sum_j \left(\frac{\sigma_j}{\sigma_j^2 + \lambda} \right)^2 a_j^2.$$

The claimed monotonicity and the feasibility criterion follow. □

Proposition 2 (Nested-prefix budget monotonicity). *If U_k is the first k columns of U_ℓ, then*

$$\mathcal{B}_{\ell,\tau}(b) \leq \mathcal{B}_{k,\tau}(b)$$

whenever the right-hand side is finite.

Proof. Embed any k-coefficient vector into the ℓ-coefficient space by appending zeros. The source vector, target residual, and source budget are unchanged. The larger feasible set can therefore only lower the infimum. □

4 Finite audit

The audit inherits the 18 rows and 1,380 shell edges used by TPC-298. Every profile and physical entry is formed over exact rational numbers before conversion to 70-digit arithmetic. For every prefix up to the available shell dimension, we record the least-squares residual and the actual source norm $c^{\mathsf{T}} M_k c$. At the first prefix whose normalized RMS is at most $\tau = 1/2$, and again at the full available prefix, we solve the KKT frontier by deterministic bisection on λ.

The weighted target is the signed minimum selected by the physical magnitude-weighted quadratic control in the preceding releases. The all-positive vector is a deliberately favorable comparison. Budget ratios are divided by the squared norm of the frozen source vector β; this is a declared finite normalization, not an asymptotic scale law.

Table 1: TPC-299 finite budget census.

quantity	count
rows / shell edges	18 / 1,380
profile cutoffs / tested prefix entries	17 / 219
weighted threshold budget $> 9 \cdot 10^{-5}\, \|\beta\|^2$	18 / 18
weighted threshold budget $> 5 \cdot 10^{-4}\, \|\beta\|^2$	15 / 18
weighted threshold budget $> 10^{-3}\, \|\beta\|^2$	14 / 18
weighted full-prefix budget $> 10^{-3}\, \|\beta\|^2$	11 / 18
positive threshold budget $< 10^{-4}\, \|\beta\|^2$	18 / 18
weighted/positive threshold budget ratio > 20	18 / 18
fixed-power credit	0

Table 2: Representative weighted threshold frontiers. The two budget columns are divided by $\|\beta\|_2^2$; full uses all available profiles.

N	H	Q	$\lvert S\rvert$	$k^{\mathrm{wt}}_{1/2}$	$\mathcal{B}_{\mathrm{wt}}/\,\|\beta\|_2^2$	$\mathcal{B}_{\mathrm{full}}/\,\|\beta\|_2^2$
128	24	9	3	2	$1.268\,10^{-3}$	$8.319\,10^{-4}$
192	32	16	5	5	$2.772\,10^{-3}$	$2.772\,10^{-3}$
512	58	60	13	9	$1.039\,10^{-2}$	$1.240\,10^{-3}$
512	58	90	17	13	$2.170\,10^{-2}$	$1.915\,10^{-3}$
384	48	70	15	13	$1.318\,10^{-1}$	$6.449\,10^{-3}$
384	52	70	15	12	$2.640\,10^{-2}$	$4.782\,10^{-3}$

The smallest weighted threshold ratio is approximately $9.545 \cdot 10^{-5}$, while the largest is approximately $1.318 \cdot 10^{-1}$. The largest positive threshold ratio is approximately $5.796 \cdot 10^{-5}$. The smallest weighted-to-positive ratio is approximately 21.849. These extrema have comfortable separation from the displayed finite thresholds.

5 Validation and interpretation

The independent checker reconstructs each physical column in source-first order rather than importing the producer's matrix. It recomputes all 219 prefix residuals, all first-threshold dimensions, and the KKT frontiers at both the threshold and full prefix. Exact fixtures include a one-dimensional active constraint with source budget exactly one, a strict budget drop after adding a coordinate, an infeasible projection radius, and the zero-source limit. Normal and optimized interpreter runs agree with empty standard error. The manuscript PDF is checked for parseability, embedded fonts, and absence of LaTeX warnings.

The finite result closes a specific ambiguity in the previous dimension ladder. The final prefix may be surjective on a finite shell, but that surjectivity does not imply a cheap native source. In the registered grid, the weighted target pays a much larger source budget than the positive control at the same target tolerance, and the excess often survives all available literal directions. This is a restricted-profile obstruction, not a statement about the unrestricted ambient source map or about every possible arithmetic profile.

6 Claim boundary and conclusion

The KKT, feasibility, ridge-path, and nesting statements are exact finite linear algebra. The numerical budget counts are certified only for the declared literal cutoff ladder, finite rows, frozen physical operator, normalization, and target controls. They do not establish uniform profile dimension, bounded source budget, arithmetic L^2, a fixed power, Gate B, or the twin-prime conjecture.

The reusable route interface is

$$\text{literal profiles} \longrightarrow A^{\mathsf{T}}U_k \longrightarrow \text{whitened ridge path} \longrightarrow \mathcal{B}_{k,\tau}(b) \longrightarrow \text{budget obstruction.}$$

The next natural test is to enlarge the shell/profile scale while preserving the source-side definition and to check whether this finite budget gap has any uniform shadow. That is an open question, not a conclusion of this paper.

Reproducibility

From the project directory run:

```
export PYTHONDONTWRITEBYTECODE=1
python -B code/tpc299_native_profile_budget_frontier_certificate.py --check
python -B experiments/tpc299_independent_checker.py
python -B experiments/tpc299_budget_stress.py
```

The Session-named Route-A/Route-B evaluator files are absent from this checkout. The theorem ledger, proof package, canonical certificate, independent replay, stress suite, PDF audit, and Bridge-B checker are the available fail-closed validation path.

References

[1] Liang Wang. Least-norm source budgets and native-ray obstruction, 2026. TPC-296 project record.

[2] Liang Wang. A literal source-profile angle and dimension ladder, 2026. TPC-298 project record.

[3] Liang Wang. A literal source-profile span for finite twin-prime shells, 2026. TPC-297 project record.

[4] Liang Wang. Source-correlation image audit for finite prime shells, 2026. TPC-295 project record.